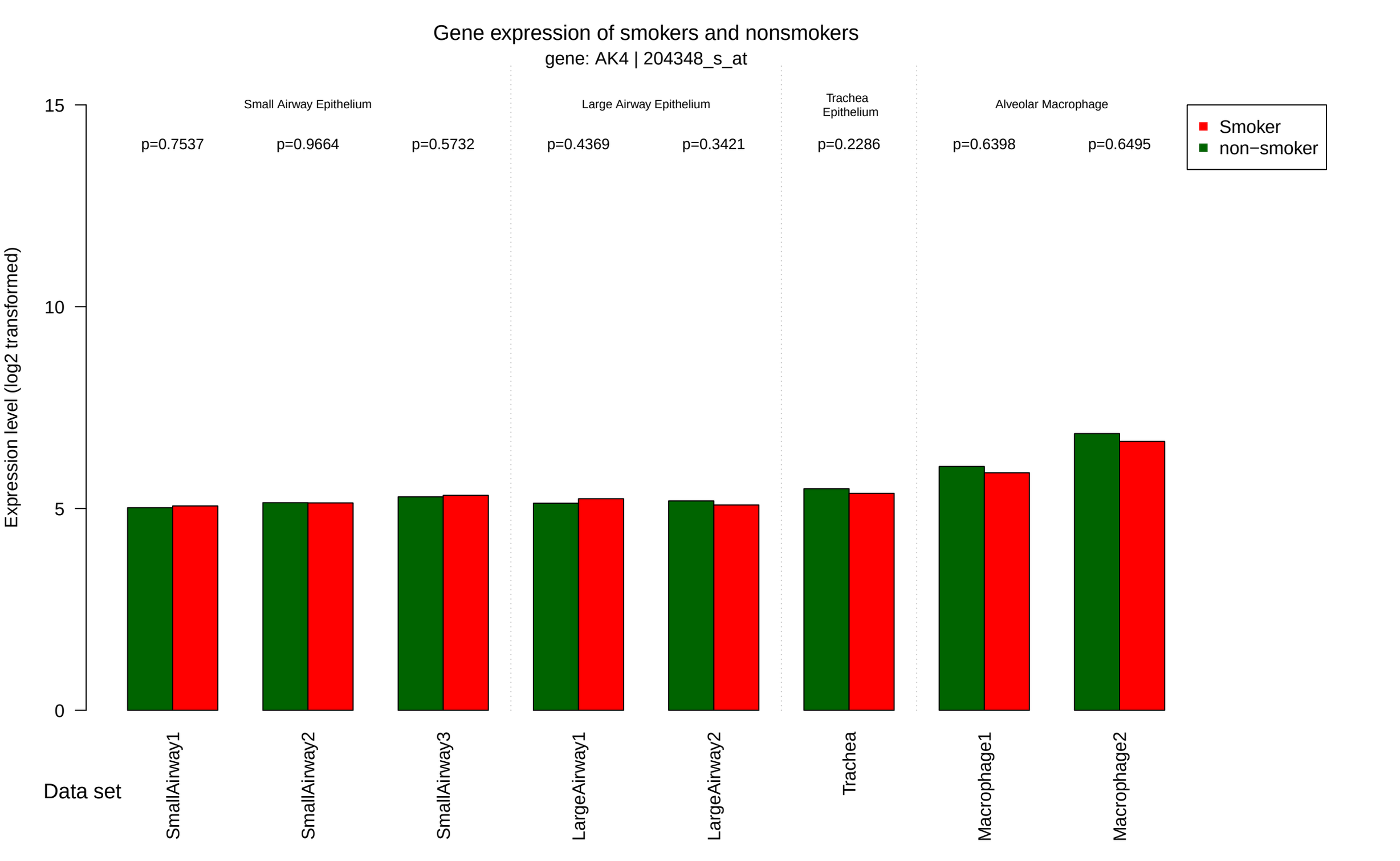
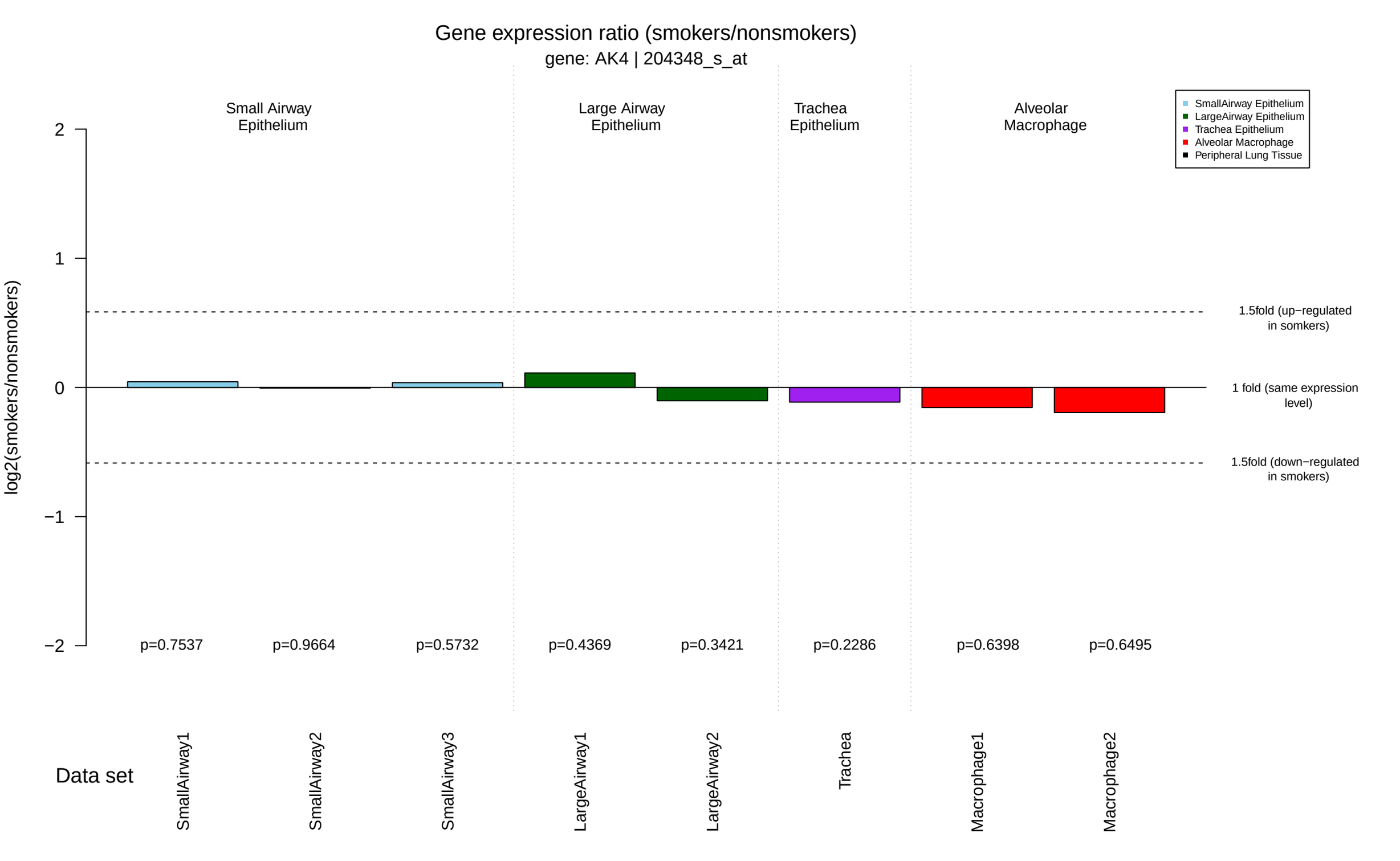
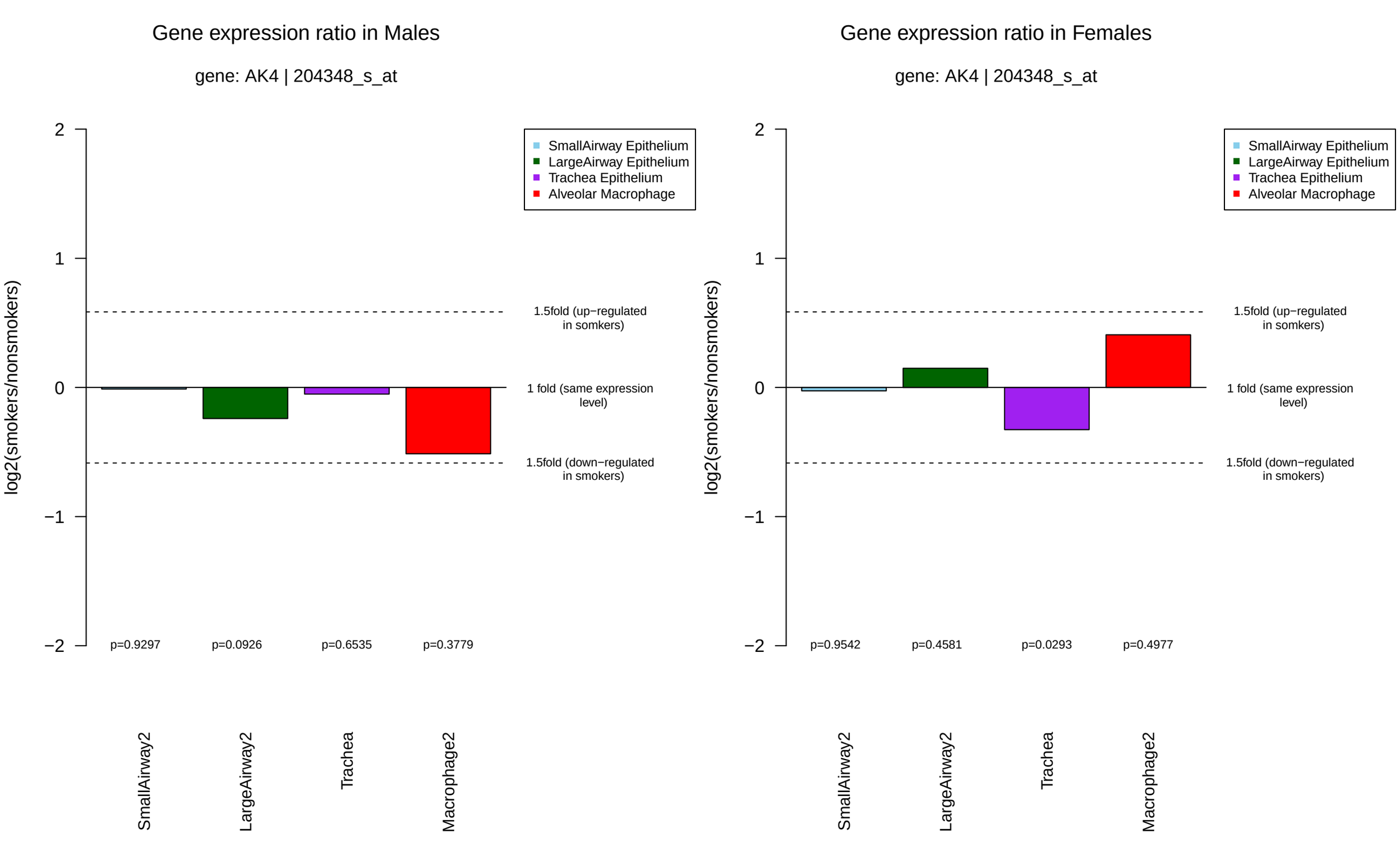
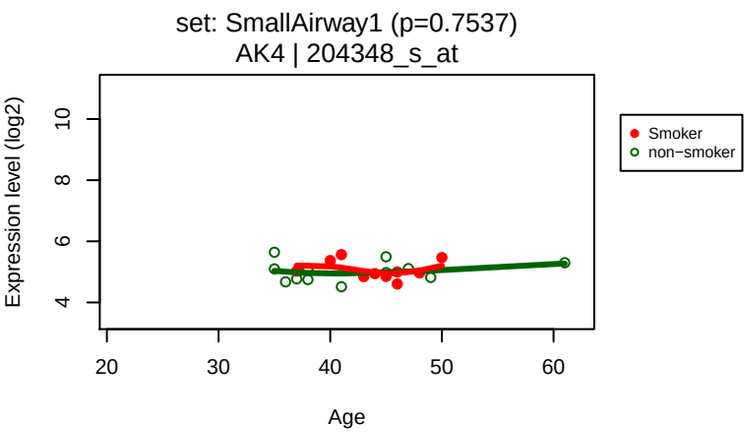
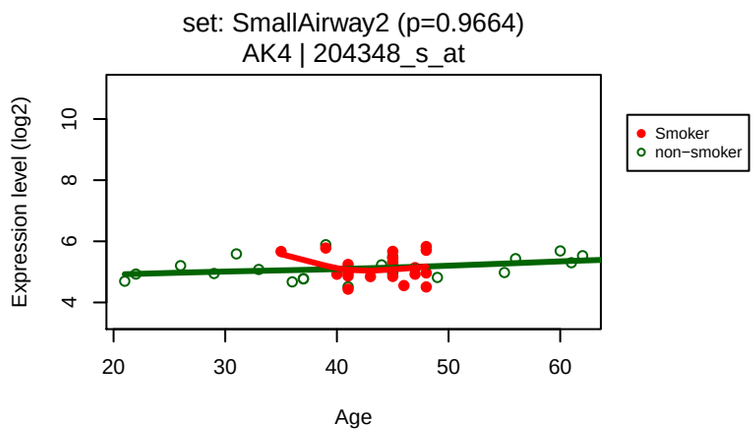
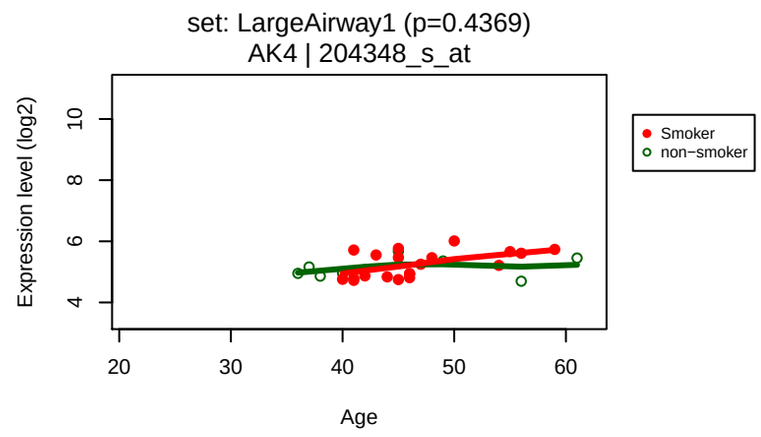

Gene expression of smokers and nonsmokers
gene: AK4 | 204348_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7537
p=0.9664
p=0.5732
p=0.4369
p=0.3421
p=0.2286
p=0.6398
p=0.6495
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

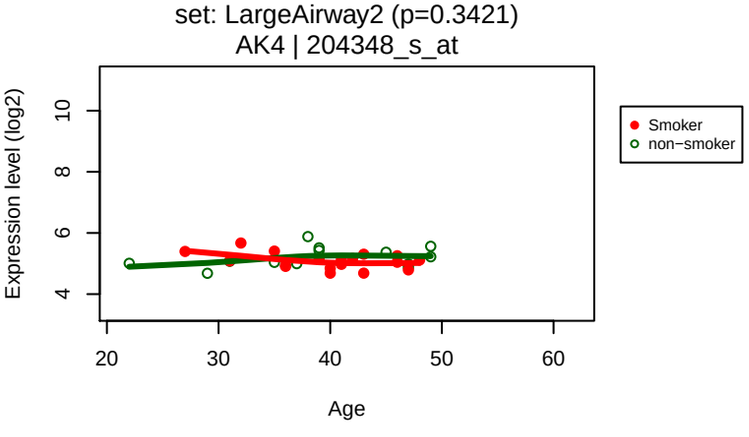

Gene expression ratio (smokers/nonsmokers)
gene: AK4 | 204348_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7537
p=0.9664
p=0.5732
p=0.4369
p=0.3421
p=0.2286
p=0.6398
p=0.6495
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

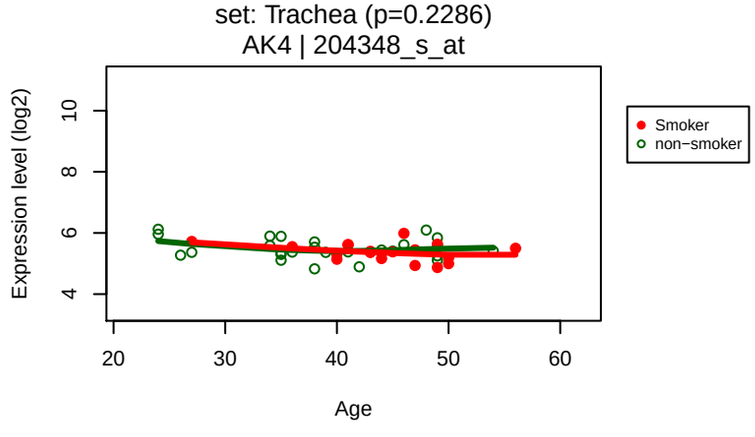

Gene expression ratio in Males
gene: AK4 | 204348_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9297
p=0.0926
p=0.6535
p=0.3779
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK4 | 204348_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9542
p=0.4581
p=0.0293
p=0.4977
SmallAirway2
LargeAirway2
Trachea
Macrophage2

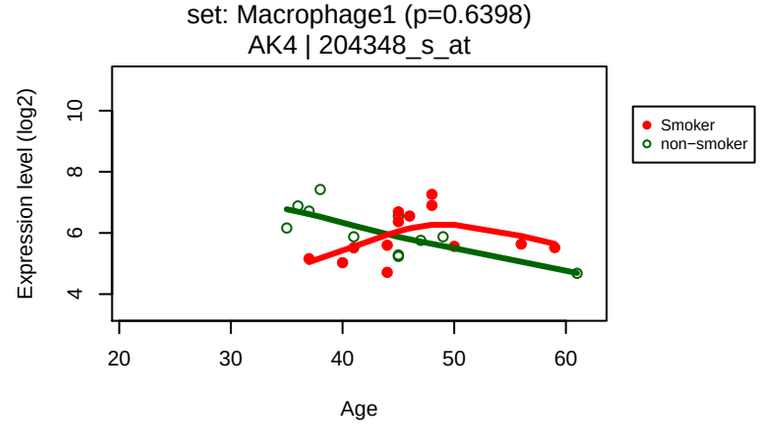

set: SmallAirway1 (p=0.7537)
AK4 | 204348_s_at
Expression level (log2)
Age
Smoker
non−smoker

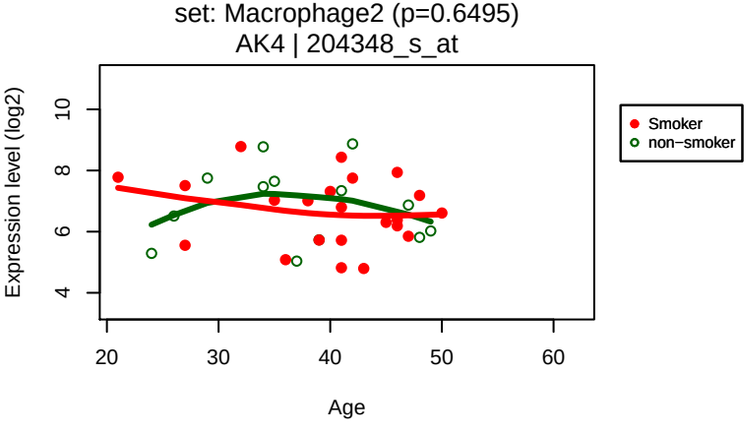

set: SmallAirway2 (p=0.9664)
AK4 | 204348_s_at
Expression level (log2)
Age
Smoker
non−smoker

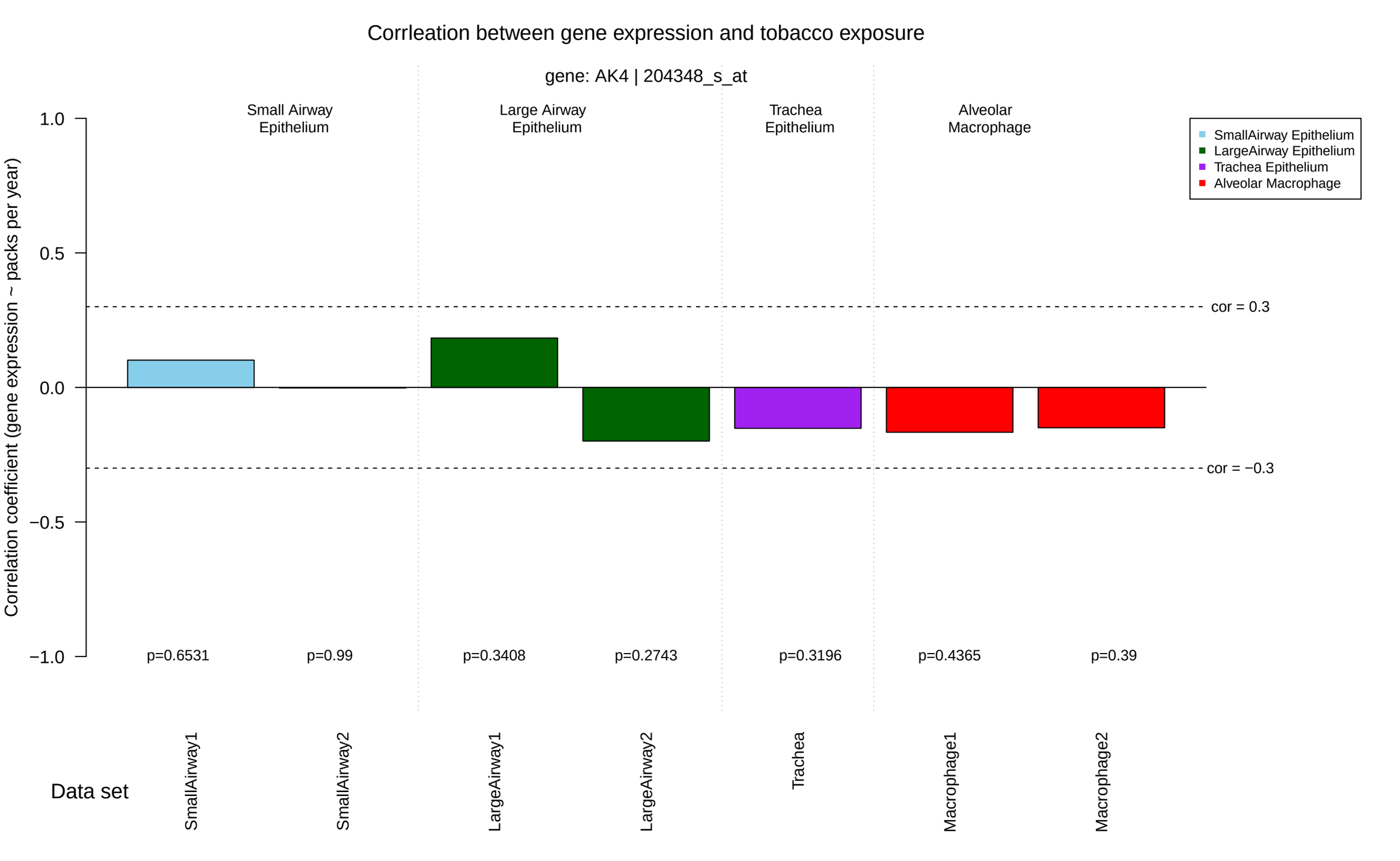

set: LargeAirway1 (p=0.4369)
AK4 | 204348_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3421)
AK4 | 204348_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2286)
AK4 | 204348_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6398)
AK4 | 204348_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6495)
AK4 | 204348_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK4 | 204348_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6531
p=0.99
p=0.3408
p=0.2743
p=0.3196
p=0.4365
p=0.39
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2